

Week 1 Progress Report

UniSupport AI: A Bounded Agentic Student Support Case System

GROUP L | BSE4104: Emerging Trends in Software Engineering

Week Ending: 4th September 2026

1. Project Overview and Context

BSE4104 requires each team to design, build, and evaluate a bounded AI-native or agentic system over an eight-week engineering cycle. In Week 1, the team's objective was to move from an open-ended problem space to a single, well-scoped, evaluable use case, and to establish the requirements, guardrails, and project infrastructure needed to begin engineering in Week 2.

The team compared two candidate proposals: a broad, multi-domain "AI University Student Assistant" and a narrower "University Student-Support Case Agent." The team selected the latter, working under the name UniSupport AI, because it is one of the brief's pre-approved use cases and its scope — answering grounded student questions, checking timetable and case status, and creating or routing support tickets — is achievable end-to-end within the eight-week timeline without relying on proprietary APIs or large datasets the team does not have access to.

2. Work Completed

The following deliverables were completed during Week 1, in line with the brief's Problem Framing and AI-Native Requirements milestone:

- Compared the two candidate proposals against the brief's scope and feasibility criteria and selected the University Student-Support Case Agent as the team's use case.
- Defined the primary end-to-end workflow: a student asks a grounded question → the agent retrieves relevant content from course and handbook documents → the agent checks timetable or case status where relevant → the agent creates or routes a support ticket → the system retains the active case in memory for the remainder of the session.
- Drafted the Project Charter, covering the problem statement, target user, current pain point, rationale for using AI, in-scope and out-of-scope boundaries, assumptions, and constraints, together with the minimum proposal statement required by the brief.
- Authored 11 user stories with acceptance criteria, covering question-answering, timetable and case-status checks, ticket creation, and session memory of the active case.
- Produced the AI Boundary Matrix, separating what the AI may do autonomously, what remains deterministic, and what is fully excluded from automation (see Section 4).
- Produced the initial context/architecture diagram, showing the user interface, the language model, the retrieval layer over course/handbook documents, the tool layer (timetable check, case-status check, ticket create/route), and the active-case memory store.
- Created the GitHub repository and set up the recommended folder structure (README, docs/, requirements/, architecture/, weekly-reports/, evaluation/, prompts/, knowledge/, src/, tests/).
- Set up the ClickUp project with a Week 1 list, assigned one task per Week 1 activity, and linked each task to the corresponding GitHub file for traceability.

3. Individual Contributions

Week 1 tasks were distributed across the team as follows:

Team Member	Task(s)	Contribution Summary
Musheija	AI Boundary Matrix; ClickUp project + task setup	Built the boundary table separating autonomous AI actions, deterministic logic, and fully out-of-scope decisions; configured the ClickUp Space, Week 1 list, and task assignments.
Okema Paul Mark	Architecture/context diagram; GitHub repository setup	Produced the initial context diagram covering the UI, LLM, RAG layer, tool calls, and memory store; created and structured the GitHub repository.
Kizito Ryan	Project Charter	Authored the 2-3 page charter covering problem statement, target user, scope boundaries, assumptions, and constraints.
Akakikunda Benitah	User stories and acceptance criteria	Drafted 11 user stories spanning the end-to-end workflow, each with 2-3 testable acceptance criteria.
Mwesigwa Duncan	Week 1 progress report	Compiled the team's Week 1 outputs into this consolidated progress report.

4. Key Engineering Decisions and Rationale

Use-case scope: the team chose the narrower support-case agent over the broader multi-domain assistant specifically to stay within the brief's scope rules and the eight-week timeline. A multi-domain assistant would have required integrating far more data sources and workflows than the team can realistically build, secure, and evaluate in the available time.

Human-control boundary: the team fixed, from Week 1 onward, that admissions, grading, disciplinary matters, and fee-related actions are never resolved autonomously by the AI. This decision satisfies the brief's human-control requirement early, before any model or agent behaviour is built, so that later engineering work is constrained by design rather than patched in afterward.

Deterministic vs. AI-driven logic: ticket routing, ticket ID generation, and timetable/case-status lookups were assigned to deterministic, non-AI logic rather than left to model inference, to keep these operationally critical steps predictable and auditable.

5. AI Boundary Matrix (Summary)

The full matrix is stored in the repository under requirements/; the categories are summarised below.

Category	Scope
AI may act autonomously	Answering grounded student questions from course/handbook documents; drafting support-ticket text for review.
Deterministic, non-AI logic	Ticket routing rules, ticket ID generation, timetable lookups, and case-status checks.
Requires human approval / fully	Admissions decisions, grading, disciplinary matters, and fee-related

out of bounds	actions are never resolved autonomously.
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6. Failures, Challenges, and Current Response

finalising which synthetic dataset fields to use for the course/handbook knowledge base — together with the team's planned resolution and target date.

7. Risks Going Into Week 2

- Availability of a realistic synthetic or public dataset for course/handbook content, timetables, and case records may constrain the scope of retrieval and tool testing.
- Coordinating five team members across GitHub and ClickUp requires consistent commit and task-update discipline to keep evidence traceable week over week.

8. Plan for Week 2

Week 2 moves the team into foundation-model engineering and prompting. Planned activities include:

- Selecting an accessible foundation model and documenting its capability and cost tradeoffs for the UniSupport AI use case.
- Building the smallest useful model-backed capability — grounded question answering — with a tested baseline before adding retrieval-augmented generation or agentic tool use.
- Beginning integration of the timetable/case-status and ticket-creation tools against the deterministic logic defined in Week 1.

9. Links

GitHub Repository: <https://github.com/Musheijaa/university-student-support-case-agent>

ClickUp Board: <https://app.clickup.com/1200440000000225/v/l/6-1200440000006918-1>